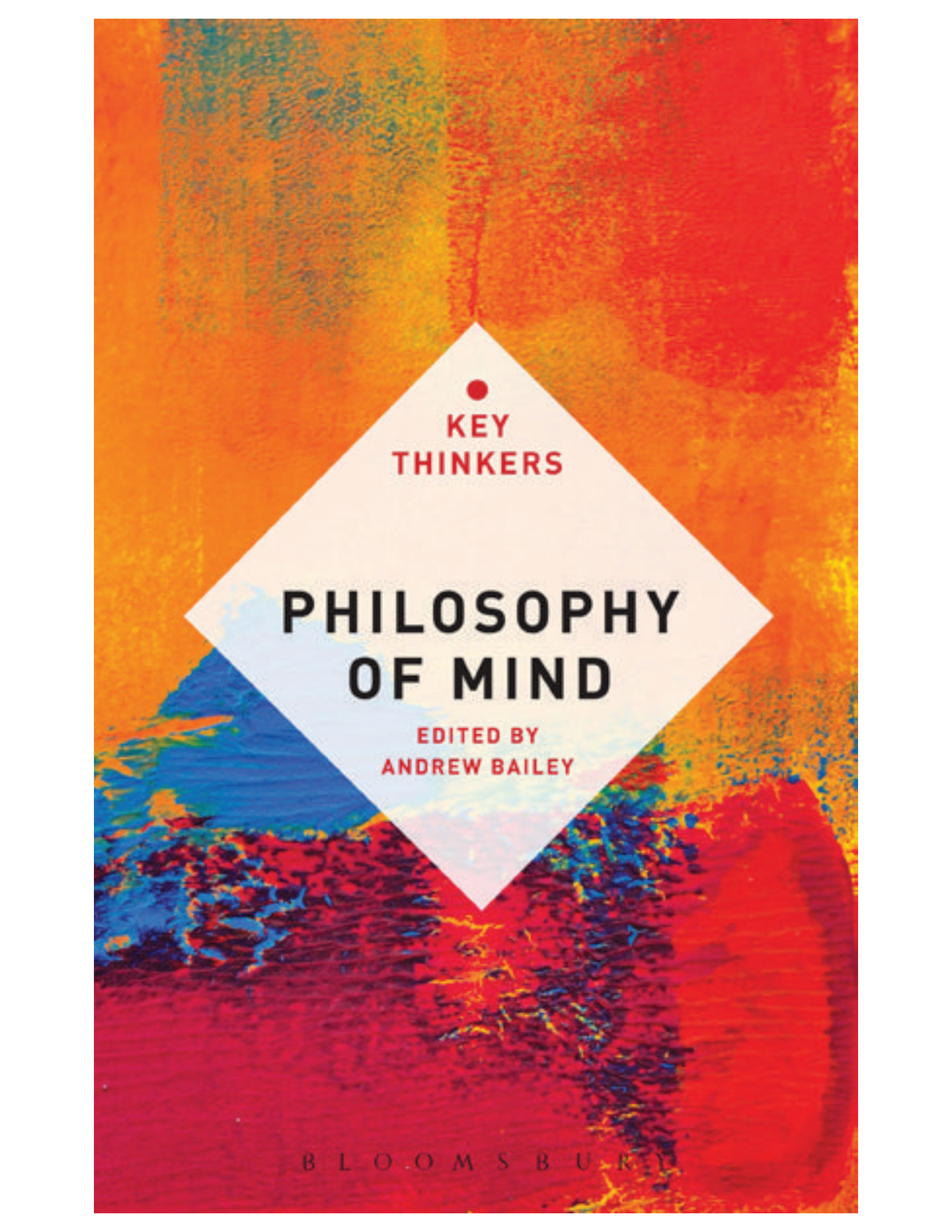


The background is an abstract, textured composition of warm colors, primarily orange, red, and yellow, with some cooler blue and green tones. A large, white, diamond-shaped area is centered on the page, serving as a container for the text. Inside the diamond, at the top, is a small red dot above the words 'KEY THINKERS' in red, uppercase letters. Below this, the title 'PHILOSOPHY OF MIND' is written in large, bold, black, uppercase letters. Underneath the title, 'EDITED BY ANDREW BAILEY' is written in red, uppercase letters. At the bottom of the page, outside the diamond, the publisher's name 'BLOOMSBURY' is written in a smaller, black, uppercase font.

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KEY
THINKERS

PHILOSOPHY OF MIND

EDITED BY
ANDREW BAILEY

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CHAPTER TWELVE

The neurophilosophies of Patricia and Paul Churchland

John Bickle

Paul M. Churchland and Patricia Smith Churchland are formative figures in late twentieth-century Anglo-American philosophy for bringing approaches and resources from the philosophy of science and neuroscience to bear on philosophy of mind. They are most famous (infamous?) for their *eliminative materialism*. But their more important long-term contributions to philosophy include proposed reformulations of the mind-body problem into first and foremost an intertheoretic reduction question and numerous other traditionally philosophical questions into *neurophilosophical* inquiries, bringing ‘connectionist’ artificial neural networks to broader philosophical consciousness and importance, and making early groundbreaking strides in neuroepistemology and neuroethics. *Philosophy of neuroscience*, a now-recognized field of academic philosophy, very much began with the Churchlands.

It is common, although mistaken, to think of the Churchlands as a single philosophical entity. Certainly, their concerns, results and *modus operandi* overlap significantly. But there are important

differences between the two, especially in philosophical and scientific focus and prose style. Hence the plural, ‘neuropsychologies’, of my title.

Eliminative materialism

Any critical assessment of the Churchlands’ corpus must begin with their eliminative materialism, so prominently a part of 1980s philosophy of mind. Eliminative materialists insist that our common-sense causal explanations of behaviour, in terms of the contents of propositional attitudes like beliefs, desires, perceptions, intentions, hopes, wishes and the like, are radically false and will be replaced by, rather than smoothly reduced to or identified with, developing neuroscientific explanations. Our intellectual history presents precedents of theories, both scientific and commonsensical, subjected to elimination – fluid theories of heat transfer, phlogiston theories of rusting and combustion, the rotating crystal sphere of the night sky, witches, demonic possession and the flat stationary earth. As we did in these cases, the eliminative materialist insists, so we’ll do with our common sense or ‘folk’ psychology: we’ll eliminate its postulated kinds and processes from our best scientific ontology. We’ll conclude that *there are no such things* as beliefs, desires, perceptions, intentions, wishes and the rest, nor the logic-like mental dynamics operating on their contents. There are only the brain states and dynamics being unravelled by neuroscience and predicted by the eliminativist to end up radically incommensurate with folk-psychological kinds.

The comparisons between folk psychology and historical eliminative theory changes are suggestive, but what arguments backed the Churchlands’ specific brand of eliminative materialism? Unfortunately, many philosophers took Paul’s argument in his then-popular textbook *Matter and Consciousness* (1983/1987) as canonical.¹ But those arguments were textbook glosses. The detailed early arguments occur in Paul’s 1981 and 1985 essays, and in Chapter 9 of Patricia’s 1986 book *Neuropsychology*. Later eliminativist arguments are in Paul’s 1989 collection *A Neurocomputational Perspective* and Patricia’s 2002 book *Brain-Wise*.

The early arguments focus on the explanatory failures and long-term developmental stagnation of folk psychology, and the

explanatory promise of some emerging paradigms in mid-1980s neuroscience. According to the Churchlands, outside of folk psychology's explanatory forte of rational, functioning, adult human behaviour, it suffers massive explanatory failures. Even in its forte, it's explanatorily silent about a lot of clearly psychological phenomena: sleep, memory, learning, intelligence differences, routine psychopathological syndromes, psychological development. The historical examples reveal that this pattern of explanatory failures is a hallmark of theories headed towards elimination.

This first eliminativist argument joins smoothly with a second. Despite its obvious explanatory failures, folk psychology hasn't developed significantly since the time of the ancient Greeks. Ancient Greek folk psychology remains contemporary folk psychology, save for a few bells and whistles slapped on over the centuries (e.g. Freud's unconscious). We read ancient Greek physics or biology, and it takes work for the scientifically trained among us to get our heads around the world picture they were groping towards. But we read ancient Greek drama, tragedy or comedy, and we're at home with the author or playwright. We understand the characters' behaviour and motivations. Why? Because our basic folk psychology is still by and large theirs. Almost nothing else in the ancient Greek worldview is still part of ours. Which explanation of this curious fact seems more reasonable? That the ancient Greeks got everything else so wrong – physics, biology, sociology – and yet got the explanation of human behaviour spot on? Or that, just like their physics, biology and social thought, the ancient Greeks probably also got psychology wrong; only, without yet having a replacement theory in hand, we contemporaries still employ their radically false psychology explanatorily, with continued poor explanatory results? The questions here are rhetorical: of course the second disjunct seems more reasonable.

A third argument was dimly present in the Churchlands' early eliminativist writings. It contains two premises. First, contemporary neuroscience is starting to succeed where folk psychology has failed. Second, neuroscience's developing explanatory resources differ so extensively from folk psychology's, in both conceptual structure and hypothesized dynamics, that future ontological identifications look hopeless. In their early writings, the Churchlands presented few neuroscientific details (aside from some stock neurology examples). This changed significantly in the mid-1980s when they discovered 'connectionist' artificial neural networks.

The Churchlands' connectionist excursion and the numerous philosophical consequences they stressed will be the focus of a later section. But they quickly emphasized its impact on their third early argument for eliminativism, especially towards developing the second premise. Paul writes:

According to [connectionism], it is activity vectors that form the most important kind of representation within the brain. And it is vector-to-vector transformations that form the most important kinds of computation. This may or may not be correct, but it does give some real substance to the earlier suggestion of the eliminative materialist . . . that the concepts of folk psychology need not capture the empirically significant states and activities of the mind. The elements of cognition, as sketched in the preceding pages, have a character unfamiliar to common sense. (1983/1987 [1987], p. 165)

From this point forward, the historical eliminations and folk psychology's stagnation didn't disappear from the Churchlands' eliminativist arguments; but these took a back seat to the differences the Churchlands noted between connectionist and folk-psychological kinds.

What became of Churchland-style eliminativism? It had its detractors from the outset and no lack of published responses. But like many philosophical fashions, the discipline more lost interest in it, rather than refuted it. Connectionism quickly fell under attack from 'classicists', and exploring its eliminativist consequences became a secondary concern to defending it as cognitive science's best paradigm (more on this below). And by the mid-1990s, philosophers of mind had consciousness as the new hot topic to grapple with, and eliminativism there seemed less plausible from the start. (More on this below, too; also see Chapter 14.) Interestingly, however, neuroscience-inspired eliminativism is making something of a comeback, at least as an argumentative foil (see Jackson, Petit and Smith 2004).

Some readers may have sensed the irony that Churchland-style eliminative materialism began as a *prediction*: about what to expect out of developing neuroscience. One wonders what it might look like if it had remained primarily a methodological recommendation about how best to develop a neuroscience of cognition, and less a

metaphysical doctrine about 'the nature of mind' (or lack thereof). In the context of a debate about explaining affect between psychologist Jim Russell and neuroscientist Jaak Panksepp, Bickle (2012b) has speculated about such an 'eliminativism with a little "e"'. Some of the distracting detours the eliminativist debate took throughout the 1980s might have been avoided.

The intertheoretic reduction reformulation of the mind-body problem

One point often lost on critics of Churchland-style eliminativism was that the account was grounded on a complete reformulation of the traditional philosophical mind-body question (see Chapter 2). The details of this reformulation actually constitute the Churchlands' most significant contribution to philosophy of mind.

It has been clear since the heyday of logical positivism that ontological disputes, like the traditional mind-body problem, are philosophical quagmires. But if those disputes can be reformulated, first and foremost into debates with clearer criteria for relevant versus irrelevant evidence and argument, yet ones which still usher in ontological consequences, so much the better. When it comes to questions about ontological relations across distinct theoretical kinds, science resolves such disputes by looking first to intertheoretic relations, principally *reduction*, between the theories postulating the kinds. Electrical and magnetic phenomena aren't identical to strictly mechanical phenomena. That follows because the developed theories of electricity and magnetism turn out not to be reducible to the theory of (classical) mechanics. This kind of strong theory autonomy, where a theory targeted for reduction ends up explaining phenomena that its potential reducing theory can't be developed to explain, implies ontological distinctness. On the other hand, light and other optical phenomena *are* electromagnetic phenomena within specific ranges of wavelengths and frequencies. That follows because optical theory reduces rather smoothly to electromagnetic theory. Temperature in gases is related to mean molecular kinetic energy of the gas's molecular constituents, but the former concept gets revised fairly significantly vis-à-vis the statistical-mechanical kinds. That follows because classical equilibrium thermodynamics

reduces to statistical mechanics and the kinetic theory of gases only in mathematical limits never actually realized empirically. And there's simply no such thing as phlogiston, only oxygen, gained and not lost in combustion and rusting. That follows because the theory of phlogiston only very bumpily reduces to the theory of oxygen chemistry. The limiting assumptions and boundary conditions required to capture the former's generalizations within the latter are so wildly counterfactual that we don't even attempt to revise the postulated kinds. We just eliminate them from our best current scientific ontology.

These considerations suggest a general strategy for reformulating the philosophical mind-body problem, into primarily an intertheoretic reduction issue, with folk psychology as the hypothesized theory-to-be-reduced and some developed future neuroscience as the theory-to-accomplish-the-reduction. As in the other scientific examples, the ontological consequences for folk psychology's kinds become secondary to and dependent upon the case's location around the intertheoretic reduction spectrum. Failure to reduce implies ontological autonomy. Smoother reductions imply ontological identities (or close-enough approximations). Bumpier reductions imply revisionary ontological conclusions for the reduced theory's kinds, and bumpier-still reductions finally yield ontological eliminations. So for a case like folk psychology-to-developed neuroscience, whose intertheoretic reduction relation is still at issue, we look to how the reduction seems to be developing. We then locate the case at that appropriate point around the intertheoretic reduction spectrum and predict ontological conclusions for the reduced theory's kinds based upon consequences drawn in historical scientific cases where the intertheoretic reduction relation turned out that way. (See Bickle 1998, Chapter 2, for a pictorial illustration of this intertheoretic reduction reformulation of the mind-body problem.)

What philosophical commitments are required to make this reformulation work? First, we need some kind of scientific realism, to get ontological commitments to science's 'theoretical' postulates. In 1980s Anglo-American philosophy, this wasn't considered problematic. Scientific realism had long replaced logical empiricism in philosophical fashion, and while van Fraassen's (1981) 'constructive empiricism' attracted some followers, it remained a minority view. Second, we need to insist that folk psychology is a theory, in an explicit, robust-enough sense for it to stand in an intertheoretic

reduction relationship to developed neuroscience. This too was not problematic at the time. 'Theory' accounts of folk psychology, and of conceptual frameworks generally, dominated Anglo-American epistemology and semantics, although Paul undertook seriously to defend the 'theory' theory (see Chapter 6 of his 1989 and Chapter 2 of Churchland and Churchland 1998). We also need a 'theory' theory of neuroscience, but few in the 1980s doubted that neuroscience was aiming at that. (Few doubt that now, in fact).

Third, and most importantly, the reformulation requires a detailed account of intertheoretic reduction. Here, Anglo-American philosophy of science obliged. Since Ernest Nagel's (1961, Chapter 11) classical account, theories of intertheoretic reduction were commonplace. The Churchlands adopted (and helped develop) Clifford Hooker's (1981) account. Hooker postulated that an 'image' of the reduced theory gets deduced within the reducing framework. His account thus eschews the need for the 'bridge laws' or 'correspondence rules' that plagued Nagel's detailed account, since the deduced image was already specified in the terms of the reducing theory, and so no principles were required to derive the image from the reducing theory. Reducing theories, which typically explain more phenomena than the theory they reduce, get related to the reduced theory with the help of various limiting assumptions and boundary conditions restricting its scope of application. When a reduction involves a false reduced theory, this falsity can be accounted for by counterfactual limiting assumptions and boundary conditions used to derive its image with the reducing theory. Both the logical structure and the explanatory power of the deduced image of the reduced theory could be compared directly to those of the actual reduced theory to obtain at least a qualitative measure of how corrective of the reduced theory a given reduction is. Hooker admitted that he could not provide a fully satisfactory formal account of 'location on the intertheoretic reduction spectrum' from less to more corrective. Construing theories set-theoretically and adapting resources from the structuralist programme in formal philosophy of science, Bickle (1998) sought to provide this missing element in the Churchland-Hooker intertheoretic reduction reformulation of the mind-body problem. The resulting account, along with the spectrum of ontological consequences justified by the scientific realist assumption, was dubbed 'new wave' psychoneural reductionism.

Criticisms followed (see Bickle 2003, Chapter 1, and Bickle 2012a for overviews). The most telling ones challenged the applicability of the general new wave account of reduction to the emerging scientific details. Bickle's response was to move increasingly towards metascience: towards detailed descriptions of the actual experimental practices that distinguished investigations in 'ruthlessly reductionistic' scientific fields (see, e.g. Bickle 2003, 2006, 2009a, 2012a). Is this new approach Churchlandian? Yes and no. Certainly, the Churchlands championed bringing increasingly detailed neuroscience into philosophical discussion. But they always did so through the lens of post-logical empiricist philosophy of science (see especially Patricia's 1986, Chapter 6) – a lens that metascience rejects (see especially Bickle 2009a).

Bechtel (2009) points out that neuroscience itself seems not to speak with one voice about what reductionism is. Cellular and molecular neuroscientists may pursue 'ruthlessly reductionistic' research programmes. But cognitive/systems/behavioural neuroscientists, addressing related cognitive phenomena, instead pursue 'mechanistic' reduction strategies: they seek nested causal mechanisms within mechanisms for the 'complete' mechanistic explanation, and they locate intraneuronal molecular mechanisms within this hierarchy of mechanistic levels, rather than claiming 'ruthless' reductions of cognitive functions exclusively to those molecular mechanisms. For a variety of methodological reasons, Bechtel (2009) sides with the mechanistic reductionists. Bickle (2012a) acknowledges multiple notions of reduction at work within contemporary neuroscience, but sides with the cellular and molecular ruthless reductionists. This ongoing debate about which brand of reductionism will win out in the end – if either – will only be resolved by continued developments in neuroscience. That prescription is certainly Churchlandian, even if their emphasis on *intertheoretic* reduction no longer directs the debate.

Connectionism

Paul Churchland's 1979 book *Scientific Realism and the Plasticity of Mind* ends by longing for a 'new kinematics' for epistemology, to replace the sentence-like representations and logic-like computations of analytic philosophy's classical model. By the early 1980s, inspired

by neuroscientific modelling of cerebellar networks, and buoyed by Paul's own tinkering with a computer-simulated crab-like creature, the Churchlands discovered the power of vectorial representations and transformations. Paul notes that his (1986) essay detailing this work was written 'with some excitement in the spring of 1984' (1989, p. xiv). That autumn, the Churchlands moved to the Department of Philosophy and the interdisciplinary Cognitive Science Ph.D. programme (soon to be the Department of Cognitive Science) at the University of California, San Diego – the home of the Parallel Distributed Processing (PDP) Research Group. There the Churchlands found the missing piece of their philosophical-scientific puzzle. They quickly became strong philosophical advocates for and tutors of this new 'connectionist' paradigm for cognitive science. Patricia published one of the earliest accounts of it in Chapter 10 of her (1986) book, the same year that the landmark two-volume collection by the PDP Research Group first appeared (Rumelhart et al 1986). Paul gave it extended discussion in the revised edition of his textbook (1983/1987, pp. 156–65), where he was quick to suggest its eliminativist consequences (mentioned above).

Explicating connectionism's virtues to philosophers was not the Churchlands' principal goal. Paul soon put the account to work in reformulating issues in the philosophy of science (see especially the essays in Part II of his 1989). He reconceived theories as points in multidimensional weight-error spaces (one dimension for each connection in the network, with the y-axis representing global error of target minus actual output) or as regions in multidimensional unit activation spaces (one dimension for each 'hidden' unit in three-layer networks). He in turn reconceived theory relations – explanation, reduction, conceptual change – as vectorial transitions in these spaces. Theoretical simplicity or parsimony, Kuhnian paradigms, explanatory unification and a variety of distinctions among types of explanations all received novel reconception in Paul's 'neurocomputational perspective'.

Were we really to take these reconceivings at face value? Churchland was explicitly affirmative:

An individual's overall theory-of-the-world, we might venture, is not a large collection or a long list of stored symbolic items. Rather, it is a specific point in that individual's synaptic weight space. It is a configuration of connection weights, a configuration

that partitions the system's activation-vector space(s) into useful divisions and subdivisions relative to the inputs typically fed the system. "Useful" here means "tends to minimize the error messages". (1989, p. 177)

His reconceivings didn't stop with the philosophy of science. By the time of his more popular 1995 book, *The Engine of Reason, the Seat of the Soul*, Churchland advocated treating consciousness, language, politics, art and neurotechnology's impact on human life all in terms of neurally inspired vectorial representations and transformations in appropriately configured state spaces, with error-reducing 'synaptic' weight-change rules driving network learning.

Unlike the attention philosophers of mind gave to the Churchlands' eliminative materialism, philosophers of science were less interested in engaging Paul's neurocomputational reconceivings. This lesser attention still strikes me as curious. This work wasn't ignored, but it certainly didn't garner the critical attention directed towards the Churchlands' previous work in philosophy of mind. Perhaps philosophers of science are more stubborn about their guiding paradigm? One principled reason for lesser attention may have been the very public fight between Classicists and Connectionists that soon erupted in cognitive science. Interestingly, the Churchlands weren't major contributors to this broader dispute. Principal Classicists included Jerry Fodor (see Chapter 9), Zenon Pylyshyn and Brian McLaughlin; principal connectionist defenders included Paul Smolensky, Terrence Horgan and John Tienson, and, more ecumenically, Andy Clark (see Chapter 13). Anecdotally, my impression is that many philosophers of science judged that the Classicists won this dispute, which could explain the surprisingly limited interest in Paul's neurocomputational reconceivings.

Perhaps more significantly, scientific interest in the kind of connectionism the Churchlands advocated also waned. Worries about biological plausibility, especially of popular and powerful 'supervised' learning algorithms, were much discussed. (Patricia, with neuroscientist co-author Terrence Sejnowski (1992), wrote one of the more vigorous defences of the biological plausibility of many connectionist components.) The advent of compartmental modelling in the mid-1990s, and the development of software that could be run on then-existing desktop computers, allowed computational neuroscientists to model the electrical conductance

capacities of patches of neuronal membrane. Individual neurons, including special dendritic, somatic and axonal properties, could be modelled as chains of interacting compartments. Networks could be constructed out of these individually modelled neurons, and so the potential for biologically plausible modelling suddenly increased dramatically. Connectionism didn't die on the vine. Its neural-like architectures, vectorial interpretations and state space geometries continued to find adherents in artificial intelligence and computational modelling. But increasingly their uses were in 'applied' areas of AI, not in the theoretical areas that interested computational neuroscientists. Intriguing and novel as Paul's 'neurocomputational' reconceptions of scientific and other concepts were, computational neuroscience soon left behind the scientific framework that informed them.

Consciousness

Consciousness studies took Anglo-American philosophy of mind by storm in the mid-1990s. The Churchlands participated avidly. Actually, their published discussions of consciousness predated its more popular return. Patricia's (1983) essay argues that neuroscientific discoveries can 'transmutate' our understanding of features of conscious experience. She appeals to neuropsychological and neurological work on phenomena like blindsight and denial syndromes (blindness, paralysis) as case studies. At the same time, Paul was taking on Thomas Nagel's well-known bat argument and Frank Jackson's equally well-known Mary argument (see P. M. Churchland 1985, 1989 Chapter 4). Nagel famously argued that objective science was unsuited to explain the subjective facts about conscious experience, such as 'what it is like to be a bat'. Jackson equally famously argued that Mary, the future vision neuroscientist who knew 'everything there is to know' about activity in the human visual system but who was confined to a room of black, white and variously shaded grey objects, would 'learn something new' with her first visual encounter with a red object, proving that conscious visual qualia (like visually experienced red) can't be identical to any brain state (for then Mary would have already known about it). Against Nagel, Paul posed a dilemma: the bat argument is an instance of the intentional fallacy – it invalidly treats someone's belief about an

object as a property of the object itself – or the argument is valid but unsound because it contains a premise about know *ability* to which Nagel could not legitimately help himself. Against Jackson, he insists that ‘knows’ is equivocated upon, sliding from a kind of ‘knowing that’ to a kind of ‘knowing how’. Patricia (1986) addresses Jackson’s insistence that Mary couldn’t know red experience based on her ‘complete’ visual neuroscience, arguing that Jackson was asking us to deliver an intuition that, in our neuroscientific ignorance, we’re in no position to deliver confidently. Paul’s (1985) reply to Jackson also contains an intriguing precursor of an idea he develops in great detail over the next decade. Mary’s knowing what it’s like to have a red visual experience while in the black-white-grey room is akin to the ability of a trained musician possessing perfect pitch to use his or her musical knowledge to ruminate on the sound of a bizarre chord (Churchland’s example was an F#9thadd13th; see his 1985) and then to pick out that chord auditorily from a range of similarly structured, equally bizarre, but slightly different chords. He appeals to Edwin Land’s (of Polaroid fame) Retinex theory of colour vision, which represents each colour visually discriminable by sighted humans as a distinct ordered triplet of reflectance frequencies at the three critical wavelengths to which the human retina’s triune cone system is tuned. Sensations-of-colour get characterized thus as three-element ‘chords’ in human neural visual processing. So if the trained musician can succeed in his or her analogous task, why not completely-neuroscientifically-knowledgable-but-as-yet-colour-inexperienced Mary in hers? She knows and has experienced the neural states associated with black, white and shades of grey. Why assume she can’t ruminate on, and then correctly pick out upon its first presentation, the triplet ‘chord’ of activations for red in her visual processing system?

By his (1995, Chapter 2) book, Paul develops some details underlying this colour vision example into a connectionism-inspired vector-coding account of all sensory representations. Tastes encode as four-element vectors, with each element representing activation levels reached in the four types of taste receptors populating the human tongue (and projected into the brainstem for further processing and integration). Each taste can thus be represented as a point in a four-dimensional taste space, depending on the activities its chemical components invoke in the four types of human taste receptors and their projection pathways. Subjective aspects of taste

experiences so difficult to describe – the experienced similarity of peach and apricot tastes, the experienced difference between quinine and table salt tastes – thus can be assessed and measured geometrically in taste space. Paul (1995) demonstrates the power of neurally realistic vector coding for human visual colour experiences, olfactory experiences (with a nicely illustrated state space comparison to canines) and even simple visually experienced facial recognition features. By his 2005 essay, 'Chimerical Colors', and armed with a more detailed neuroscientific account of colour visual processing, he argues that a vector-coding account yields novel predictions about 'chimerical', 'impossible' colours. He even provides the colour slides so readers can 'savor' these novel colour experiences themselves – experiences predicted directly from the computational neurobiology.

As the consciousness craze swept 1990s philosophy of mind, the Churchlands also challenged philosophers who held out against the reducibility of conscious experience to neuroscience. John Searle's famous insistence, that conscious mental states are 'caused by and realized in' neural states, but not thereby reducible to them, reminds the Churchlands of Betty Crocker's mistaken insistence that microwave ovens agitate water molecules in food causing friction that in turn heats the food (see Churchland and Churchland 1998, Chapter 8). Actually, the increased molecular agitation of water molecules *constitutes* the heated food, with a similar lesson to be urged on Searle for the neural states underlying consciousness. They also insisted that the neurobiologically based vector-coding account of sensory qualia offers an at least empirically motivated (if not yet confirmed) solution to Chalmers's (1996) famous Hard Problem of Consciousness (see Chapter 14). Coupling that account with recurrency (feedback connections) in networks yields a potential account of short-term memory and top-down cognitive effects on sensory conscious experience. 'Inverted experiential spectra' are thus not only conceptually possible, but explicable in detail. (For their arguments for all three of these points, see Churchland and Churchland 1998, Chapter 11). Reflection on some well-known examples from the history of science exposes popular dualist lessons drawn from 'the conceivability of philosophical zombies' (Chalmers 1996) to be nothing more than arguments from ignorance – ones which even ongoing early work in neuroscience already disarm (see P. S. Churchland 1996).

Many (myself included) found the Churchlands' vector-coding account of sensory experiences intriguing. It seemed to offer a plausible story about how sensory experiences within a modality resembled and differed from one another. But more reductive neuroscientific work on consciousness was emerging from cellular and molecular neuroscience (Bickle 2003, Chapter 4). Cortical microstimulation of single cerebral columns throughout primate visual cortex seemed to induce phenomenological experiences (Bickle and Ellis 2005). Genetic engineering of single amino acid residues on various protein subunits of 'fast' GABA_A inhibitory receptors produced mutant mice with quite selective impairments on behavioural measures of various features of phenomenological consciousness (Bickle 2007). It began to look like the Churchlands' account of sensory consciousness was not as reductionistic as ongoing neuroscientific research.

And yet the qualitative features of conscious experience still seemed maddeningly elusive! Why does induced activity in those microstimulated motion-detecting V4 neurons look *like that*? (Like what? Like the way an object moving at a particular speed at a particular angle across my visual field looks). Why not instead *like that* (the colour aquamarine)? Whether one rests content with the Churchlands at the brain's vector-coding level or digs further down into the cellular and molecular mechanisms, answers to these questions about sensory experiences still seemed elusive. Perhaps we still have to be satisfied with neurobiologist William Newsome's answer? Reflecting on his own and others' microstimulation results in rhesus monkeys, Newsome remarks:

I believe the nature of internal experience matters for our understanding of nervous system function. . . . Even if I could explain a monkey's behavior on our task in its entirety (in neural terms), I would not be satisfied unless I knew whether microstimulation in MT actually causes the monkey to see motion. . . . For the time being . . . I suspect we must feel our way towards these ambitious goals from the bottom up (1997, pp. 65–7)

It's important to notice that the Churchlands' neurally inspired vector-coding account of sensory consciousness is less reductionistic than is the approach recommended by this prominent neurophysiologist.

Neuroethics and neuroepistemology, up to the present day

Paul's vector-coding work towards a neuroepistemology continues in his latest book, *Plato's Camera* (2012). Here, his emphasis is on neural details of learning, of both dynamically fast and ploddingly slow varieties. Biological plausibility has become more of a concern. He argues that neural mechanisms of Hebbian learning overcome some of the neurobiological implausibility of supervised learning in connectionist networks. The learning brain is groping towards a picture of the landscape of reality. He urges us to take the camera metaphor seriously. Those pictures are composed of high-dimensional maps of regions of reality's landscape. He offers a new defence of scientific realism and increasingly detailed accounts of the semantics of natural language, and the roles of natural language and cultural institutions in shaping the brain's pictures of reality. The basic neurocomputational framework from his studies (1989) and (1995) remains in place.

The Churchlands' contributions to the burgeoning field of neuroethics have also been significant. Paul ended his (1989) collection with a cryptic short essay suggesting a realist view of moral facts and moral knowledge, along similar lines to his scientific realism. The 'neuro' component of his view is nicely summed up the essay's two closing sentences: 'For in fact we do have an organ for understanding and recognizing moral facts. It is called the brain' (1989, p. 303). In his (1995) book, these cryptic remarks get cashed out explicitly in his vector-coding approach. Moral learning is skill learning and moral knowledge is a kind of knowing-how, namely, knowing how to navigate the complex world of human social relationships. Paul does not shy away from normative ethics, comparing his resulting neurocomputational account to Aristotle's virtue ethics (1995, pp. 149–50) and posing an answer to the moral sceptic's 'why be moral?' query: 'because they are easily the most important skills you will ever learn' (1995, p. 294). He even ventures into applied ethics, discussing abortion, criminal policy – including California's then-recently passed 'three strikes' law (1995, pp. 305–14) – and the need to keep public secondary education religiously neutral (2005, Chapter 5).

But it was Patricia who really took up the call to neuroethics. Starting with her discussion of free will in her (2002, Chapter 5)

book, Patricia likewise puts to use the familiar vector-coding neurocomputational account. But she also delves deeply into the neurochemistry and molecular biology of motivation and decision-making, especially the neurotransmitters and hormones involved in these neural circuits. These details become increasingly prominent in her (2009) essay, and even more so in her latest book *Braintrust* (2012). Inference to the *best decision*, rather than to the best explanation, becomes her overarching neuroethical theme. Aristotle, pragmatism and evolutionary theory join with computational neuroscience and cellular and molecular detail, to produce a tour de force about ‘what neuroscience tells us about morality’.

The jury of professional opinion is still out on both of the Churchlands’ recent books. Patricia’s (2012) has seen positive reviews in both *Science* and *Nature*, and numerous reviews in the popular media. Reviews of Paul’s (2012) have been mostly in academic journals. One worry has been a lack of updated scientific references, especially for an epistemology that purports to be neurally guided.

Philosophical styles and personalities

One cannot conclude a chapter on the Churchlands without some comment on their philosophical styles and personalities. Both are personally charismatic in different ways, which attract both devoted followers and harsh critics. Their early rhetoric was revolutionary in words and tone – they were leading an effort to transform a tired, old academic discipline, philosophy, with input from one of the hottest young sciences, neuroscience. Nothing in the former would ever be the same again. Paul was particularly gifted at rhetorical flourishes. In one of my personal favourites, he’s reflecting on ‘new opportunities’ offered by maturing neuroscience and the need to leave ‘old frameworks’ like folk psychology behind:

Our eyes are little different than a baboon’s or a chimpanzee’s, but our perceptual knowledge is profoundly superior to theirs. Our motor systems are little different from those of any other primate, but our practical capacities and intentional actions encompass universes quite closed to them. The main difference

lies in the dramatically superior conceptual frameworks we have evolved epigenetically, and not without misadventure, over the course of the last 500,000 years.

If we have come this far, must the journey end here? Manifestly not. The long awakening is potentially endless. The human spirit will continue its breathtaking adventure of self-reconstruction, and its perceptual and motor capacities will continue to develop as an integral part of its self-reconstruction. But only if we try hard to see new opportunities, and only if we work hard at leaving old frameworks behind. (1989, p. 279)

To which Jerry Fodor, at his curmudgeonly best, replied: 'An endless awakening sounds like not all that much fun, come to think of it. I, for one, am simply unable to self-construct until I've had my morning coffee' (1988, p. 198). For those who think philosophical writing is dull, this full Churchland-Fodor exchange is a panacea: an excellent example of two premier stylists going toe-to-toe, with no philosophical, scientific or rhetorical holds barred.

Paul's philosophical-cum-scientific creativity is nicely on display towards the end of his (1981) essay, where he takes us 'beyond folk psychology' to a future of distinct brains outfitted with artificial commissures connecting them, with the speed and fluidity with which our own biological corpus callosum connects neurons across our left and right cerebral hemispheres. How will such people understand and conceive of others with whom they share opened channels? 'In roughly the same fashion that your right hemisphere "understands" and "conceives of" your left hemisphere: intimately and efficiently, but not propositionally!' (1989, p. 21) For Paul, linguistic descriptions of philosophical points aren't always sufficient. I've already mentioned the colour slides published with his (2005) essay, allowing readers to experience impossible, chimerical colours predicted directly from the neuroscience. His (1995) book comes equipped with a plastic stereoscope, and stereoscopic photographs of aerial views of the New York City skyline and the planets Jupiter, Saturn and Mars, against the stars of the constellation Virgo, photographed one month apart. These give readers the opportunity to *experience* reality from vastly different visual perspectives. His (1979) book contains simple illustrations of an exercise in perceptual plasticity that enable readers to experience

our solar system with the ecliptic as the horizon – and finally to be ‘home in one’s universe’ visually as well as conceptually.

Patricia’s philosophical style is less showy, more direct, but no less effective. She’s a master at the unexpected humorous one-liner, inserted into otherwise direct argument. For example, concerning the in-principle possibility for psychology to develop a complete functionalist cognitive theory without help from neuroscience, she writes: ‘I don’t know whether this is true, and I have no sense of how to assess the claim. My guess is that it shares a flaw with many other philosophical thought-experiments: too much thought and not enough experiment’ (1986, pp. 362–3). She can be devastatingly blunt, down to the level of individual word choice (e.g. ‘pander’):

In the mid-seventies I discovered that my patience with most mainstream philosophy had run out. What had instead begun to seem promising was the new wave in philosophical method, which ceased to pander to “ordinary language” and which began in earnest to reverse the anti-scientific bias typical of “linguistic analysis”. (1986, p. ix)

Her revolutionary rhetoric can match Paul’s, and even routinely employs Kuhnian phrasing:

Paradigms rarely fall with decisive refutation; rather, they become enfeebled and slowly lose adherents. . . . [M]any of us sense that working within “the grand old paradigm” is not very rewarding. . . . There are remarkable new developments in cognitive neurobiology which encourage us to think that a new and encompassing paradigm is emerging. (1989, p. 546)

The Churchlands’ personalities are likewise larger-than-life and come out nicely in a well-written New Yorker profile (Macfarquhar 2007). Paul and Patricia narrate numerous stories about their shared roads to philosophical and scientific stardom. It’s salubrious to see that the revolutionary rhetoric that inspired the early days of neurophilosophy still motivates them:

One afternoon recently, Paul says, he was home making dinner when Pat burst in the door, having come straight from a frustrating faculty meeting. “She said, ‘Paul, don’t speak to me, my serotonin

levels have hit bottom, my brain is awash in glucocorticoids, my blood vessels are full of adrenaline, and if it weren't for my endogenous opiates I'd have driven the car into a tree on the way home. My dopamine levels need lifting. Pour me a Chardonnay, and I'll be down in a minute.'" Paul and Pat have noticed that it is not just they who talk this way—their students now talk of psychopharmacology as comfortably as of food. (2007, p. 68)

(The ruthless reductionist in me asks, 'What are these "car", "tree", "home", and "Chardonnay" of which you speak, Pat?' But set internal neurophilosophical squabbles aside.)

What will be the legacy of Paul and Patricia Churchland? Historically, their legacy is secure. Their eliminative materialism will continue to be archived in future philosophy textbooks. Their neurocomputational perspective may have been based ultimately on dated computational neuroscience, but the neuroepistemology and neuroethics it spawned in their later work will continue to have impact, both in academic and in popular circles. Few contemporary philosophers can truly be said to have invented an entire field of the discipline. Admittedly, current philosophy of neuroscience looks quite different than the Churchlands' first envisioned: contrast Patricia's (1986) book with the chapters in Bickle's (2009b) edited volume. Much of the early revolutionary rhetoric is now gone. There's little dispute that analytic philosophy's armchair crowd proved far more resistant than the Churchlands expected. But today's philosophers of neuroscience are mostly people who've done graduate work, and even earned graduate degrees, in the neurosciences. And all this work traces directly back to the days when Paul and Patricia Churchland started 'a cautious paddling at the available edges of neuroscience', quickly found themselves 'venturing further and further from shore', and finally setting 'full sail' (P. S. Churchland 1986, p. ix). It was a pied piper's call for a lot of us, and many of us are still trying to kindle that same excitement in the budding philosopher-neuroscientists we encounter.

Note

- 1 Since I'll emphasize throughout this chapter some differences between the Churchlands' neurophilosophies, I'll sometimes have to resort to less formal first-name usage.

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